

LOCAL AUTOMORPHISMS IS A LIE GROUP

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ABSTRACT. In this paper, we prove the conjecture on the structure of local automorphism proposed by Ayupov, Elduque, and Kudaybergenov (*J. Pure Appl. Algebra*, 2023). We prove that the set of all local automorphisms of a finite-dimensional algebra forms a Lie group, whose Lie algebra is precisely the space of local derivations.

Key words: local automorphism

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1. INTRODUCTION

Derivations are classical tools in the study of algebraic structures. They describe infinitesimal symmetries and are closely related to automorphisms, extensions, deformations, and nilpotency. Many variants and generalizations of derivations have also been studied. For a broad account of these topics, we refer the reader to the recent survey of Kaygorodov [15] and the references therein.

A linear map $\Delta: A \rightarrow A$ is called a *local derivation* if, for every $x \in A$, there exists a derivation D_x of A , depending on x , such that

$$\Delta(x) = D_x(x).$$

Local derivations were introduced in operator algebra theory by Kadison [14] and were further studied by Larson and Sourour [17] and Johnson [13].

A linear map $\Phi: A \rightarrow A$ is called a *local automorphism* if, for every $x \in A$, there exists an automorphism φ_x of A , depending on x , such that

$$\Phi(x) = \varphi_x(x).$$

The set of all local automorphisms of A is denoted by $\text{LocAut}(A)$. Local automorphisms were introduced in operator algebra theory by Larson and Sourour [17] and were further studied by Brešar and Šemrl [6, 7].

These developments lead naturally to a structural question raised by Ayupov, Elduque, and Kudaybergenov [1]: Let A be a finite-dimensional algebra over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Is $\text{LocAut}(A)$ a Lie group, and is $\text{LocDer}(A)$ a Lie algebra under the commutator bracket? If so, is the Lie algebra of $\text{LocAut}(A)$ naturally isomorphic to $\text{LocDer}(A)$?

Recent work by Hao and Chen proved that the set of all local derivations of a finite-dimensional algebra forms a Lie algebra [10], leaving the following two conjectures open:

Conjecture 1. Let A be a finite-dimensional algebra over $\mathbb{R}$ or $\mathbb{C}$. Then $\text{LocAut}(A)$ is a Lie group.

Conjecture 2. Let A be a finite-dimensional algebra over $\mathbb{R}$ or $\mathbb{C}$. If $\text{LocAut}(A)$ is a Lie group, then its Lie algebra is naturally isomorphic to the space of local derivations:

$$\text{Lie}(\text{LocAut}(A)) \cong \text{LocDer}(A).$$

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In [10], Hao and Chen utilize the reflexive hull to unify the concepts of local and almost inner derivations. The algebraic reflexive hull was introduced by Hadwin [8] and was subsequently developed for spaces of operators and linear interpolation by Larson [16]. A general abstract framework was later given by Hadwin [9]. For arbitrary sets and semigroups of operators, algebraic reflexive hulls were studied by Bračič [4], and a recent account of the finite-dimensional theory can be found in [5]. Indeed, if A is a finite-dimensional algebra and L is a finite-dimensional Lie algebra, then

$$\text{LocDer}(A) = \text{Ref}_a(\text{Der}(A)), \quad \text{AID}(L) = \text{Der}(L) \cap \text{Ref}_a(\text{Inn}(L)).$$

The perspective of the reflexive hull provides profound insights into the study of local derivations, almost inner derivations, and local automorphisms. In fact, if E possesses a certain algebraic structure, its reflexive hull $\text{Ref}_a(E)$ typically inherits a similar algebraic structure. This is precisely the underlying philosophy of the reflexive hull: for instance, since $\text{Der}(A)$ naturally carries a Lie algebra structure, it is highly natural to conjecture that $\text{Ref}_a(\text{Der}(A))$ also admits a Lie algebra structure.

Guided by this perspective, we can express the space of local automorphisms via the reflexive hull, namely $\text{LocAut}(A) = \text{Ref}_a(\text{Aut}(A))$. Since $\text{Aut}(A)$ naturally carries a Lie group structure, it is natural and highly plausible to conjecture that $\text{LocAut}(A)$ forms a Lie group. More generally, one might be tempted to conjecture that if $E \subseteq \text{GL}(V)$ is a Lie subgroup, then $\text{Ref}_a(E)$ is also a Lie subgroup of $\text{GL}(V)$. However, this turns out to be false. In fact, we require a more refined geometric structure of $\text{Aut}(A)$. Over the complex, we view $\text{Aut}(A)$ as a complex algebraic group, whereas over the real, we consider its structure as a Nash group. This structural characterization enables our aforementioned strategy to succeed. Together with the fact that both complex algebraic groups and Nash groups are Lie groups, we ultimately complete the proof of the conjecture.

This paper is organized as follows. In Section 2, we treat the complex and real cases separately by exploiting the structure of $\text{Aut}(A)$ as a complex algebraic group over $\mathbb{C}$ and as a Nash group over $\mathbb{R}$. This allows us to prove that the set of all local automorphisms forms a Lie group in both settings, thereby proving Conjecture 1. In Section 3, we show that the Lie algebra of this newly established Lie group coincides precisely with the space of local derivations, thus proving Conjecture 2.

Throughout this paper, unless otherwise stated, all vector spaces are finite-dimensional, and by an algebra we mean such a vector space equipped with a single bilinear multiplication.

2. LIE GROUP STRUCTURE OF LOCAL AUTOMORPHISMS

The purpose of this section is to prove Conjecture 1 in a sharper form. Although the conjecture only asks whether $\text{LocAut}(A)$ carries a Lie group structure, our conclusions retain substantially more geometric information. Over $\mathbb{C}$, the natural structure is algebraic: we prove that $\text{LocAut}(A)$ is a Zariski closed algebraic subgroup, and hence a closed embedded complex Lie subgroup. Over $\mathbb{R}$, the appropriate framework is semialgebraic and Nash geometry: we prove that $\text{LocAut}(A)$ is a semialgebraic subgroup, and hence a closed Nash submanifold and a closed embedded real Lie subgroup. The two cases require different geometric arguments, but both are based on the same algebraic-reflexivity framework.

We begin by recalling the algebraic reflexive hull of a family of linear operators.

Definition 2.1. [16] Let V be a vector space and let $E \subseteq \text{End}(V)$. The *algebraic reflexive cover*, also called the *algebraic reflexive hull*, of E is defined by

$$\text{Ref}_a(E) = \{f \in \text{End}(V) \mid f(v) \in E(v) \text{ for all } v \in V\},$$

where $E(v) = \{g(v) \mid g \in E\}$.

Remark 2.2. Let A be a finite-dimensional algebra. Then

$$\text{LocDer}(A) = \text{Ref}_a(\text{Der}(A)), \quad \text{LocAut}(A) = \text{Ref}_a(\text{Aut}(A)).$$

Thus local derivations and local automorphisms can be studied uniformly through the algebraic reflexive hulls of $\text{Der}(A)$ and $\text{Aut}(A)$, respectively.

This follows from the fact that the algebraic reflexive hull preserves the relevant structure.

Proposition 2.3. [4] *Let V be a vector space.*

- (1) *If $\mathcal{S} \subseteq \text{End}(V)$ is a semigroup, then $\text{Ref}_a(\mathcal{S})$ is a semigroup.*
- (2) *If V is finite-dimensional and $G \leq \text{GL}(V)$, then $\text{Ref}_a(G)$ is a subgroup of $\text{GL}(V)$.*

Applying Proposition 2.3(2) to the automorphism group $\text{Aut}(A) \leq \text{GL}(A)$ immediately yields the following preliminary consequence.

Corollary 2.4. *Let A be a finite-dimensional algebra over an arbitrary field. Then $\text{LocAut}(A)$ is a subgroup of $\text{GL}(A)$.*

Having established that $\text{LocAut}(A)$ is a subgroup of $\text{GL}(A)$, we now turn to its geometric structure. Our goal is to prove that $\text{LocAut}(A)$ is a Lie subgroup. A natural approach would be to deduce this from the identity $\text{LocAut}(A) = \text{Ref}_a(\text{Aut}(A))$ and the fact that $\text{Aut}(A)$ is a Lie subgroup. However, as the following example demonstrates, the reflexive hull of an embedded Lie subgroup is not necessarily an embedded Lie subgroup.

Example 2.5. Let

$$\alpha \in \mathbb{R} \setminus \mathbb{Q}, \quad V = \mathbb{C}^2 \oplus \mathbb{C}^2, \quad d_z = \text{diag}(e^z, e^{\alpha z}),$$

and set

$$E = \left\{ \text{diag}(A, d_z) \mid A \in \text{GL}_2(\mathbb{C}), z \in \mathbb{C}, \det A = e^{iz} \right\}.$$

Then E is a closed embedded complex Lie subgroup of $\text{GL}_4(\mathbb{C})$, whereas

$$\text{Ref}_a(E) = \{ \text{diag}(P, d_z) \mid P \in \text{GL}_2(\mathbb{C}), z \in \mathbb{C} \}$$

is not closed and hence is not an embedded Lie subgroup of $\text{GL}_4(\mathbb{C})$.

Therefore, we require a more refined geometric structure of $\text{Aut}(A)$. We first consider the complex case. Our argument relies on constructible sets and the fact that a constructible subgroup of an algebraic group is Zariski closed. We therefore recall the necessary terminology and results from [3, 11, 12].

2.1. The complex case.

Definition 2.6. [11] Let k be an algebraically closed field and let $\mathbb{A}_k^n = k^n$ denote the n -dimensional affine space over k . For a subset $S \subseteq k[x_1, \dots, x_n]$, set

$$V(S) = \{a \in \mathbb{A}_k^n \mid f(a) = 0 \text{ for all } f \in S\}.$$

The *Zariski topology* on $\mathbb{A}_k^n$ is the topology whose closed subsets are precisely the subsets of the form $V(S)$.

Definition 2.7. [12] An *affine algebraic variety* over k is a Zariski closed subset

$$X \subseteq \mathbb{A}_k^n$$

for some $n \geq 0$, equipped with the induced Zariski topology. Equivalently, X is the common zero set of a family of polynomials.

Definition 2.8. [3] Let X be an affine algebraic variety. A subset $Y \subseteq X$ is called *locally closed* if

$$Y = U \cap Z$$

for some Zariski open subset $U \subseteq X$ and some Zariski closed subset $Z \subseteq X$. A subset $C \subseteq X$ is called *constructible* if it is a finite union of locally closed subsets.

Remark 2.9. The constructible subsets of an affine algebraic variety X form a Boolean algebra. In particular, they are closed under finite unions, finite intersections, and complements relative to X . Consequently, if $A, B \subseteq X$ are constructible, then

$$A \setminus B = A \cap (X \setminus B)$$

is constructible.

The converse implication is false in general. Namely, from the assumptions that $A \subseteq X$ is constructible and $A \setminus B$ is constructible, one cannot conclude that B is constructible. However, if in addition $B \subseteq A$, then

$$B = A \setminus (A \setminus B),$$

and hence B is constructible.

Definition 2.10. [12] Let $X \subseteq \mathbb{A}_k^m$ and $Y \subseteq \mathbb{A}_k^n$ be affine algebraic varieties over an algebraically closed field k . A map

$$\varphi : X \longrightarrow Y$$

is called a *morphism of affine algebraic varieties* if there exist polynomials

$$f_1, \dots, f_n \in k[x_1, \dots, x_m]$$

such that

$$\varphi(x) = (f_1(x), \dots, f_n(x)) \quad \text{for all } x \in X.$$

Lemma 2.11. [12] Let $f : X \rightarrow Y$ be a morphism of affine algebraic varieties over an algebraically closed field k . If $C \subseteq X$ is constructible, then $f(C) \subseteq Y$ is constructible.

In particular, if X and Y are affine algebraic varieties and $C \subseteq X \times Y$ is constructible, then its image under the coordinate projection

$$\text{pr}_1 : X \times Y \longrightarrow X$$

is constructible.

Definition 2.12. [12] A *complex algebraic group* is a complex affine algebraic variety G , together with a group structure on G , such that the multiplication map

$$m : G \times G \longrightarrow G, \quad (g, h) \longmapsto gh,$$

and the inverse map

$$\iota : G \longrightarrow G, \quad g \longmapsto g^{-1},$$

are morphisms of algebraic varieties.

Definition 2.13. [12] A subgroup $H \leq G$ is called a *Zariski closed subgroup*, or a *closed algebraic subgroup*, if it is Zariski closed as a subset of G .

Remark 2.14. Let A be a finite-dimensional complex algebra. The underlying additive group of A is a complex algebraic group, and $\text{GL}(A)$ is a complex algebraic group. Moreover,

$$\text{Aut}(A) = \{g \in \text{GL}(A) \mid g(xy) = g(x)g(y) \text{ for all } x, y \in A\}$$

is a Zariski closed subgroup of $\text{GL}(A)$.

Lemma 2.15. [3] Let G be a complex algebraic group and let $H \leq G$. If H is constructible as a subset of G , then H is Zariski closed in G .

We are now ready to prove the main result of the complex case. The argument shows, more generally, that the algebraic reflexive hull of any Zariski closed algebraic subgroup of $\mathrm{GL}(V)$ is again Zariski closed.

Theorem 2.16. *Let V be a finite-dimensional complex vector space and let $G \leq \mathrm{GL}(V)$ be a Zariski closed algebraic subgroup. Then $\mathrm{Ref}_a(G)$ is a Zariski closed algebraic subgroup of $\mathrm{GL}(V)$.*

Proof. By Proposition 2.3, $H := \mathrm{Ref}_a(G)$ is a subgroup of $\mathrm{GL}(V)$. It remains to prove that H is constructible.

Define $R = \{(T, x) \in \mathrm{GL}(V) \times V \mid \exists g \in G \text{ such that } Tx = gx\}$. Consider

$$\tilde{R} = \{(T, x, g) \in \mathrm{GL}(V) \times V \times G \mid Tx = gx\}.$$

The condition $Tx = gx$ is given by polynomial equations in the matrix entries of T , the coordinates of x , and the matrix entries of g . Hence $\tilde{R}$ is Zariski closed in $\mathrm{GL}(V) \times V \times G$. Let

$$p : \mathrm{GL}(V) \times V \times G \longrightarrow \mathrm{GL}(V) \times V$$

be the projection onto the first two factors. Then $R = p(\tilde{R})$. By Lemma 2.11, R is constructible in $\mathrm{GL}(V) \times V$.

By Definition 2.1, $T \in H$ if and only if $(T, x) \in R$ for all $x \in V$. Taking the complement, this is equivalent to saying that $T \notin H$ if and only if there exists $x \in V$ such that $(T, x) \notin R$. In terms of sets, this yields the identity

$$\mathrm{GL}(V) \setminus H = \mathrm{pr}_1((\mathrm{GL}(V) \times V) \setminus R),$$

where $\mathrm{pr}_1 : \mathrm{GL}(V) \times V \rightarrow \mathrm{GL}(V)$ denotes the first projection. Since the family of constructible sets is closed under complementation, $(\mathrm{GL}(V) \times V) \setminus R$ is constructible. Applying Lemma 2.11 once again, we conclude that its image under pr_1 , namely $\mathrm{GL}(V) \setminus H$, is also constructible.

Constructible subsets are closed under complements, so H is constructible. Since H is a subgroup of the algebraic group $\mathrm{GL}(V)$, Lemma 2.15 implies that H is Zariski closed in $\mathrm{GL}(V)$. Consequently $\mathrm{Ref}_a(G) = H$ is a Zariski closed algebraic subgroup of $\mathrm{GL}(V)$. $\square$

Since $\mathrm{Aut}(A)$ is a Zariski closed algebraic subgroup of $\mathrm{GL}(A)$, and every algebraic group is a Lie group, we immediately obtain the following corollary.

Corollary 2.17. *Let A be a finite-dimensional complex algebra. Then $\mathrm{LocAut}(A) \subseteq \mathrm{GL}(A)$ is a Zariski closed algebraic subgroup. In particular, $\mathrm{LocAut}(A)$ is a closed embedded complex Lie subgroup of $\mathrm{GL}(A)$.*

2.2. The real case. The real case requires a different geometric framework. The real Zariski topology is too coarse to capture the Euclidean topology relevant to Lie group structures, and therefore Zariski closedness is not the natural primary conclusion in this setting. Instead, we work in the semialgebraic category and then use Nash geometry.

We begin by recalling the basic notions of semialgebraic geometry.

Definition 2.18. [2] A subset $X \subseteq \mathbb{R}^n$ is called *semialgebraic* if it is a finite union of subsets of the form

$$\{x \in \mathbb{R}^n \mid f_1(x) = \cdots = f_r(x) = 0, \quad g_1(x) > 0, \dots, g_s(x) > 0\},$$

where $f_i, g_j \in \mathbb{R}[x_1, \dots, x_n]$. A map $f : X \rightarrow Y$ between semialgebraic subsets of Euclidean spaces is called *semialgebraic* if its graph $\Gamma_f = \{(x, f(x)) \mid x \in X\}$ is semialgebraic.

Remark 2.19. Let $X \subseteq \mathbb{R}^n$ be a semialgebraic set. Then semialgebraic subsets of X are stable under finite unions, finite intersections and complements in X . In particular, if $A, B \subseteq X$ are semialgebraic, then $A \setminus B$ is semialgebraic. Let $B \subseteq A$, if both A and $A \setminus B$ are semialgebraic, then so is B .

The following result is the semialgebraic counterpart of Lemma 2.11 used in the complex case.

Lemma 2.20. [2] *Let $f : X \rightarrow Y$ be a semialgebraic map between semialgebraic sets. If $C \subseteq X$ is semialgebraic, then $f(C) \subseteq Y$ is semialgebraic.*

In particular, if X and Y are semialgebraic sets and $C \subseteq X \times Y$ is semialgebraic, then its image under the coordinate projection

$$\text{pr}_1 : X \times Y \longrightarrow X$$

is semialgebraic.

Definition 2.21. [19] A *Nash structure* on a topological space M is an element

$$n \in \{-\infty, 0, 1, 2, \dots\}$$

together with a set $\mathcal{N}$ satisfying the following properties:

- (a) The set $\mathcal{N}$ is contained in $\mathcal{N}(\mathbb{R}^n, M)$, where $\mathcal{N}(\mathbb{R}^n, M)$ denotes the set of all triples

$$(\phi, U, U')$$

such that U is an open semialgebraic subset of $\mathbb{R}^n$, U' is an open subset of M , and

$$\phi : U \longrightarrow U'$$

is a homeomorphism.

- (b) Every two elements

$$(\phi_1, U_1, U'_1), \quad (\phi_2, U_2, U'_2)$$

of $\mathcal{N}$ are Nash compatible, namely, the homeomorphism

$$\phi_2^{-1} \circ \phi_1 : \phi_1^{-1}(U'_1 \cap U'_2) \longrightarrow \phi_2^{-1}(U'_1 \cap U'_2)$$

has a semialgebraic domain and codomain, and is semialgebraic and smooth.

- (c) There are finitely many elements

$$(\phi_i, U_i, U'_i) \in \mathcal{N}, \quad i = 1, 2, \dots, r,$$

where $r \geq 0$, such that

$$M = U'_1 \cup U'_2 \cup \dots \cup U'_r.$$

- (d) For every element of $\mathcal{N}(\mathbb{R}^n, M)$, if it is Nash compatible with all elements of $\mathcal{N}$, then it is an element of $\mathcal{N}$.

- (e) If M is empty, then $n = -\infty$.

Definition 2.22. [19] A *Nash manifold* is a smooth manifold M equipped with a Nash structure on it. A continuous map $f : M \rightarrow N$ between Nash manifolds is called a *Nash map* if, for every pair of Nash charts

$$\phi : U \longrightarrow \phi(U), \quad \psi : V \longrightarrow \psi(V),$$

the set

$$\phi(U \cap f^{-1}(V))$$

is semialgebraic, and the coordinate representation

$$\psi \circ f \circ \phi^{-1} : \phi(U \cap f^{-1}(V)) \longrightarrow \psi(V)$$

is smooth and semialgebraic.

Definition 2.23. [19] A *Nash group* is a group G equipped with a Nash manifold structure such that the multiplication map

$$m : G \times G \longrightarrow G, \quad (g, h) \longmapsto gh,$$

and the inversion map

$$\iota : G \longrightarrow G, \quad g \longmapsto g^{-1},$$

are Nash maps.

A subgroup $H \leq G$ of a Nash group is called *semialgebraic* if it is a semialgebraic subset of the Nash manifold G .

Remark 2.24. Let A be a finite-dimensional real algebra. The underlying additive group of A is a Nash group, and $\mathrm{GL}(A)$ is a Nash group. Moreover, $\mathrm{Aut}(A)$ is a semialgebraic subgroup of $\mathrm{GL}(A)$.

We are now ready to establish the main result in the real case. As in the complex case, the essential step is to show that the algebraic reflexive hull remains in the relevant geometric category.

Theorem 2.25. *Let V be a finite-dimensional real vector space and let $G \leq \mathrm{GL}(V)$ be a semialgebraic subgroup. Then $\mathrm{Ref}_a(G)$ is a semialgebraic subgroup of $\mathrm{GL}(V)$.*

Proof. By Proposition 2.3,

$$H := \mathrm{Ref}_a(G)$$

is a subgroup of $\mathrm{GL}(V)$. It remains to prove that H is semialgebraic.

Fix a basis of V , so that V is identified with $\mathbb{R}^n$ and $\mathrm{GL}(V)$ with $\mathrm{GL}_n(\mathbb{R})$. Define

$$R = \{(T, x) \in \mathrm{GL}(V) \times V \mid \text{there exists } g \in G \text{ such that } Tx = gx\}.$$

Consider

$$\tilde{R} = \{(T, x, g) \in \mathrm{GL}(V) \times V \times G \mid Tx = gx\}.$$

The condition $Tx = gx$ is given by polynomial equations in the matrix entries of T and g and the coordinates of x . Since G is semialgebraic, it follows that $\tilde{R}$ is semialgebraic in $\mathrm{GL}(V) \times V \times G$. Let

$$p : \mathrm{GL}(V) \times V \times G \longrightarrow \mathrm{GL}(V) \times V$$

be the projection onto the first two factors. Then

$$R = p(\tilde{R}).$$

By Lemma 2.20, R is semialgebraic in $\mathrm{GL}(V) \times V$.

By Definition 2.1, $T \in H$ if and only if $(T, x) \in R$ for every $x \in V$. Taking complements, this is equivalent to saying that $T \notin H$ if and only if there exists $x \in V$ such that $(T, x) \notin R$. Hence

$$\mathrm{GL}(V) \setminus H = \mathrm{pr}_1((\mathrm{GL}(V) \times V) \setminus R),$$

where

$$\mathrm{pr}_1 : \mathrm{GL}(V) \times V \longrightarrow \mathrm{GL}(V)$$

is the first projection. Since semialgebraic sets are closed under complements, $(\mathrm{GL}(V) \times V) \setminus R$ is semialgebraic. Applying Lemma 2.20 once again, we conclude that $\mathrm{GL}(V) \setminus H$ is semialgebraic. Therefore H is semialgebraic.

Thus $H = \mathrm{Ref}_a(G)$ is a semialgebraic subgroup of the Nash group $\mathrm{GL}(V)$. $\square$

Proposition 2.26. [19] *Every semialgebraic subgroup of a Nash group G is a closed Nash subgroup of G .*

Combining Proposition 2.26 with Theorem 2.25, we obtain the following consequence.

Corollary 2.27. *Let V be a finite-dimensional real vector space and let $G \leq \mathrm{GL}(V)$ be a closed Nash subgroup. Then $\mathrm{Ref}_a(G)$ is a closed Nash subgroup of $\mathrm{GL}(V)$.*

Since $\mathrm{Aut}(A)$ is a semialgebraic subgroup of $\mathrm{GL}(A)$. By Proposition 2.26, $\mathrm{Aut}(A)$ is a closed Nash subgroup, and every closed Nash subgroup is a closed embedded real Lie subgroup. Combining this fact with the preceding corollary, we immediately obtain the following corollary.

Corollary 2.28. *Let A be a finite-dimensional real algebra. Then $\mathrm{LocAut}(A) \subseteq \mathrm{GL}(A)$ is a closed Nash subgroup. In particular, $\mathrm{LocAut}(A)$ is a closed embedded real Lie subgroup of $\mathrm{GL}(A)$.*

3. THE LIE ALGEBRA OF THE LOCAL AUTOMORPHISM GROUP

Having established that $\mathrm{LocAut}(A)$ is an embedded Lie subgroup of $\mathrm{GL}(A)$, we now identify its Lie algebra and thereby prove Conjecture 2. We first recall the notion of a local flow.

Definition 3.1. [18] Let Y be a smooth vector field on a smooth manifold M . A *local flow* of Y is a smooth map

$$\Phi : \mathcal{D} \longrightarrow M,$$

where $\mathcal{D} \subseteq \mathbb{R} \times M$ is an open subset such that, for every $p \in M$,

$$I_p := \{t \in \mathbb{R} \mid (t, p) \in \mathcal{D}\}$$

is an open interval containing 0. The map Φ satisfies

$$\Phi(0, p) = p$$

and

$$\Phi(t, \Phi(s, p)) = \Phi(t + s, p)$$

whenever both sides are defined. Moreover, its infinitesimal generator is Y , that is,

$$Y_p = \left. \frac{d}{dt} \right|_{t=0} \Phi(t, p).$$

The term “local” refers to the fact that, for each $p \in M$, the flow is defined only for t in the possibly proper interval I_p . If $\mathcal{D} = \mathbb{R} \times M$, then Φ is called a global flow.

For every $p \in M$, the curve

$$t \longmapsto \Phi(t, p), \quad t \in I_p,$$

is the integral curve of Y through p ; equivalently,

$$\frac{d}{dt} \Phi(t, p) = Y_{\Phi(t, p)}$$

for every $t \in I_p$.

We now apply this local flow property to the orbits of linear Lie group actions.

Lemma 3.2. *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let V be a finite-dimensional $\mathbb{F}$ -vector space, and let $G \leq \mathrm{GL}_{\mathbb{F}}(V)$ be an $\mathbb{F}$ -Lie subgroup with Lie algebra $\mathfrak{g}$. Let $O = Gx$ be an orbit, equipped with its standard smooth structure for which the orbit map*

$$G/G_x \longrightarrow O, \quad gG_x \longmapsto gx,$$

is a diffeomorphism. Then:

(1) *O is an immersed submanifold of V , and*

$$T_y O = \mathfrak{g}y = \{Xy \mid X \in \mathfrak{g}\}, \quad y \in O.$$

(2) Let Y be a smooth vector field on V satisfying

$$Y(y) \in T_y O, \quad y \in O.$$

If Φ is the local flow of Y , then, for every $y \in O$, there exists $\delta_y > 0$ such that

$$\Phi(t, y) \in O$$

with $|t| < \delta_y$, for which $\Phi(t, y)$ is defined.

Proof. For $y \in O$, consider the orbit map

$$\rho_y : G \longrightarrow V, \quad \rho_y(g) = gy.$$

Here $(d\rho_y)_I$ denotes its differential at the identity $I \in G$. By the standard orbit theorem for smooth Lie group actions, $O = Gy = Gx$, equipped with its standard smooth structure, is an immersed submanifold of V , and

$$T_y O = \text{Im}(d\rho_y)_I.$$

For $X \in \mathfrak{g}$, we have

$$(d\rho_y)_I(X) = \left. \frac{d}{dt} \right|_{t=0} \rho_y(\exp(tX)) = \left. \frac{d}{dt} \right|_{t=0} \exp(tX)y = Xy.$$

Consequently,

$$T_y O = \{Xy \mid X \in \mathfrak{g}\} = \mathfrak{g}y,$$

which proves (1).

For (2), let $i : O \longrightarrow V$ denote the inclusion. Since

$$Y(y) \in T_y O \quad \text{for every } y \in O,$$

the restriction of Y to O determines a smooth vector field Y_O on O , characterized by

$$di_y((Y_O)_y) = Y(y), \quad y \in O.$$

Fix $y \in O$. Let $\gamma : (-\varepsilon, \varepsilon) \longrightarrow O$ be the integral curve of Y_O through y , so that $\gamma(0) = y$. Then $i \circ \gamma$ is an integral curve of the ambient vector field Y . Indeed,

$$\left. \frac{d}{dt} \right|_{t=0} (i \circ \gamma)(t) = di_{\gamma(t)}(\gamma'(t)) = di_{\gamma(t)}((Y_O)_{\gamma(t)}) = Y(i(\gamma(t))).$$

On the other hand, by the definition of a local flow, the curve $t \longmapsto \Phi(t, y)$ is an integral curve of Y through the same initial point y . By uniqueness of integral curves, for some sufficiently small $\delta > 0$, we have

$$\Phi(t, y) = i(\gamma(t))$$

for all $|t| < \delta$ for which the flow is defined. Since $\gamma(t) \in O$, it follows that

$$\Phi(t, y) \in O$$

for all sufficiently small t . This proves (2). $\square$

Based on the preceding lemma, we are now ready to prove Conjecture 2.

Theorem 3.3. *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let V be a finite-dimensional $\mathbb{F}$ -vector space, and let $G \leq \text{GL}_{\mathbb{F}}(V)$ be an $\mathbb{F}$ -Lie subgroup with Lie algebra $\mathfrak{g}$. Assume that $\text{Ref}_a(G)$ is an embedded $\mathbb{F}$ -Lie subgroup of $\text{GL}_{\mathbb{F}}(V)$. Then*

$$\text{Lie}(\text{Ref}_a(G)) = \text{Ref}_a(\mathfrak{g})$$

as subspaces of $\text{End}_{\mathbb{F}}(V)$.

Proof. Put $H = \text{Ref}_a(G)$, and let $X \in \text{Lie}(H)$. Choose a smooth curve $T(t) \in H$ such that $T(0) = I$ and $T'(0) = X$. For every $v \in V$, the curve $T(t)v$ lies in the orbit Gv . By Lemma 3.2, its velocity at $t = 0$ lies in $T_v(Gv)$. Since

$$T_v(Gv) = \{Yv \mid Y \in \mathfrak{g}\} = \mathfrak{g}v,$$

we get $Xv \in \mathfrak{g}v$ for all v . Hence $X \in \text{Ref}_a(\mathfrak{g})$.

Conversely, let $X \in \text{Ref}_a(\mathfrak{g})$. Then the linear vector field $w \mapsto Xw$ is tangent to every G^0 -orbit, where G^0 is the identity component of G . By Lemma 3.2, its flow e^{tX} locally preserves each G^0 -orbit. Therefore, for fixed $v \in V$, the set

$$I_v = \{t \in \mathbb{R} \mid e^{tX}v \in G^0v\}$$

is nonempty, open, and has open complement. Since $\mathbb{R}$ is connected, $I_v = \mathbb{R}$. Thus $e^{tX}v \in G^0v \subseteq Gv$ for all v and all $t \in \mathbb{R}$. Hence $e^{tX} \in H$ for all t .

Since H is embedded in $\text{GL}_{\mathbb{F}}(V)$, the ambient smooth curve $t \mapsto e^{tX}$ is an H -valued smooth curve. Its derivative at $t = 0$ is X , so $X \in \text{Lie}(H)$. This proves the equality.

When $\mathbb{F} = \mathbb{C}$, the above argument is applied to the underlying real Lie groups. The two sides are complex vector spaces, and H is a complex Lie subgroup; hence the resulting real equality is the desired complex equality. $\square$

Corollary 3.4. *Let A be a finite-dimensional algebra over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Then*

$$\text{Lie}(\text{LocAut}(A)) = \text{LocDer}(A)$$

as Lie subalgebras of $\text{End}(A)$.

Proof. Let $G = \text{Aut}(A)$. By Corollary 2.17 in the complex case and Corollary 2.28 in the real case,

$$\text{LocAut}(A) = \text{Ref}_a(G)$$

is an embedded Lie subgroup of $\text{GL}(A)$. It is well known that $\text{Lie}(G) = \text{Der}(A)$. Therefore Theorem 3.3 gives

$$\text{Lie}(\text{LocAut}(A)) = \text{Lie}(\text{Ref}_a(G)) = \text{Ref}_a(\text{Lie}(G)) = \text{Ref}_a(\text{Der}(A)).$$

But $\text{Ref}_a(\text{Der}(A))$ is exactly the definition of $\text{LocDer}(A)$. Thus

$$\text{Lie}(\text{LocAut}(A)) = \text{LocDer}(A)$$

inside $\text{End}(A)$. $\square$

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CONFLICTS OF INTEREST

The authors declare that they have no conflicts of interest.

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